11 · Ad exposure is self-selected — the true effect (instrumental variables · CausalPy)

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> **Runs in the `legacy` environment (pymc<6): `make env-legacy`, kernel `cmp-legacy`.**

The business decision. People who see our ads convert more — but the customers who *end up* seeing our ads are also the ones already leaning toward us (they browse, they search, they get retargeted). So **exposure is self-selected**, and a plain regression of sales on exposure is biased **upward**: it credits the ad for enthusiasm that was already there. We want the *true* causal effect of an extra exposure so we can set budgets and frequency caps that don't overspend.

1.0.1 The problem in one word: endogeneity

When the treatment (ad exposure) is **correlated with the hidden drivers of the outcome** (a customer's unobserved intent), we say exposure is **endogenous**, and no amount of controlling for what we *measured* fixes it — the confounder (intent) is unmeasured. Notebook 05's adjustment tools are powerless here.

1.0.2 The fix: an instrument

An **instrumental variable (IV)** is a third variable Z that nudges the treatment *without* affecting the outcome any other way. Here Z is a randomized **encouragement** — e.g. a serving-priority lottery that makes some customers *more likely* to be shown the ad. Because Z is randomized, the slice of exposure it drives is as-good-as-random, uncontaminated by intent. IV isolates that clean slice and reads the effect off it. A valid instrument needs three things:

- **Relevance** — Z actually moves exposure (this one we *can* test: the **first-stage F-statistic**; F < 10 is a "weak instrument" danger sign).
- **Exclusion** — Z affects sales *only through* exposure (the lottery itself sells nothing). Untestable from data — defended on design, and stress-tested in Step 6.
- **Exogeneity** — Z is independent of the hidden confounder (guaranteed here by randomization).

1.0.3 What IV recovers: a LATE

IV does **not** give the average effect over everyone. It gives the **LATE** (*Local Average Treatment Effect*) — the effect *for the "compliers,"* the customers whose exposure the encouragement actually changed. Always-exposed loyalists and never-exposed sceptics are silent in an IV estimate; report the scope accordingly.

> **On real data.** IV needs a genuine instrument, which is the hard part. Good marketing instruments: randomized **encouragement / intent-to-treat** designs (ship the nudge to a random half), ad-auction or delivery randomness, or a **cost shifter** for the price-elasticity problem in notebook 03. If you can randomize the *encouragement* (not the exposure itself), you have a clean instrument.

7-step contract.

1.1 2 · Simulate a ground truth

The **true** exposure->sales effect is **€15**. An unobserved `engagement` drives both exposure and sales (the confounding), so naive OLS overstates it. A randomized `encouragement` (the instrument) nudges exposure without touching sales directly.

```
TRUE exposure->sales effect €15  ·  first-stage: exposure 56%->77% (shift +21%),
F = 156
```

	encouragement	ad_exposure	sales
0	1.0	0.0	63.748992
1	1.0	1.0	64.636639
2	1.0	1.0	46.507627
3	0.0	1.0	75.599770
4	0.0	1.0	70.577754

1.2 3 · Identify — a valid instrument, and what IV recovers

Exposure is **endogenous** ($\mathrm{Cov}(\text{exposure}, \varepsilon) \neq 0$), so OLS is biased. A valid instrument Z (encouragement) needs three things:

- **Relevance** — Z moves exposure. *Testable* (the first-stage F above; F<10 = weak).
- **Exclusion** — Z affects sales *only through* exposure. *Untestable* — defended on design (a serving lottery sells nothing by itself). Stress-tested in step 6.
- **Exogeneity** — Z independent of the confounder. Holds by randomization.

What IV identifies — LATE (Imbens–Angrist): under monotonicity (no defiers), IV recovers the effect for **compliers** — customers the instrument moves — not everyone. CausalPy fits a Bayesian IV: a joint model of (exposure, sales) with correlated errors, *estimating* the endogeneity ρ rather than assuming it away.

1.3 4 · Estimate — OLS, reduced form, and Bayesian IV

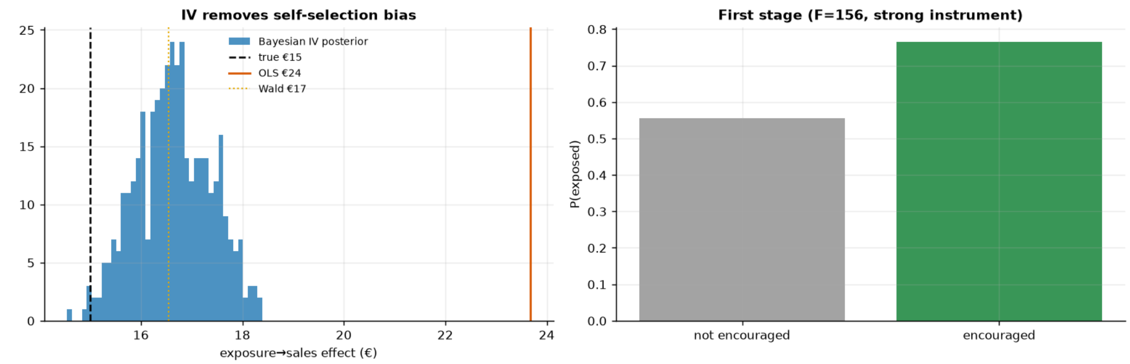
Three regressions tell the whole story: **OLS** (biased), the **reduced form** (encouragement->sales, the numerator of the IV ratio), and **Bayesian IV** (the causal estimate).

```
OLS (biased)      €23.7
Reduced form (Z->Y) €3.5  ÷  first stage (Z->D) 0.21  =  Wald/IV ratio €16.5
Bayesian IV         €16.6  (true €15)  90% CI [€15.5, €17.8]
```

1.4 5 · Validate — IV removes the self-selection bias

Naive OLS should sit well *above* the truth; the Wald ratio and the Bayesian IV should both land near €15. The gap between OLS and IV **is** the self-selection bias the instrument removes.

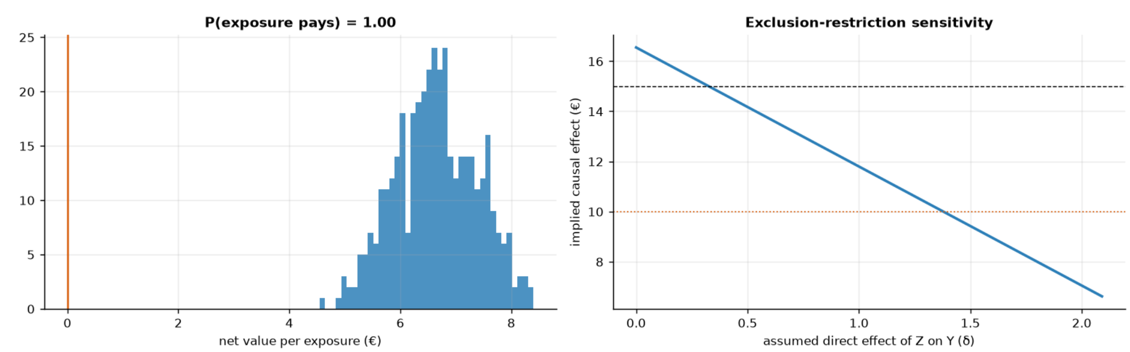
```
self-selection bias removed: OLS €23.7 -> IV €16.6 (true €15)
```



1.5 6 · Decide, in euros — and the exclusion-restriction stress test

Budget on the *causal* per-exposure value (IV), not the inflated OLS. But the exclusion restriction is untestable, so we ask: **what if the encouragement had a small direct effect on sales** (say the lottery email itself advertises)? We subtract a hypothetical direct effect δ from the reduced form and watch the implied causal estimate move — the honest bound on how much a leak would distort the number.

```
Causal net €6.6/exposure  ·  P(pays) 1.00 -> BUY exposure (to the complier
margin)
Exclusion stress: the call survives unless the instrument's direct effect
exceeds delta~1.4 (out of a €3.5 reduced form). If you trusted OLS (€24) you'd
over-spend and set frequency caps too high.
```



1.6 7 · Caveats

- **LATE, not ATE.** IV speaks only for **compliers** — customers the encouragement moves. Always-exposed loyalists and never-exposed skeptics may respond differently; report the scope.
- **Weak instruments are dangerous.** A small first-stage (low F) inflates variance and bias. We checked F above; a weak instrument is worse than none.
- **Exclusion is untestable.** We stress-tested it in step 6; defend it on design, not data.
- **Monotonicity (no defiers).** IV's LATE interpretation assumes no one is pushed *away* from exposure by encouragement — usually plausible, worth stating.